Does scrolling more mean feeling worse?

Executive Summary, ACDT Project 1: Revisit "Tackling the Conventional Wisdom"
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Interactive data story (scroll, filters, calculators): dhkim060225-web.github.io/group32-sns-mental-health
Code: github.com/dhkim060225-web/group32-sns-mental-health

The conventional wisdom and our question

Many people believe that the more time students spend on social media, the worse they feel and the worse they do in class. We set out to test whether passive social media use lowers grades. The only usable dataset had no GPA and no measure of passive versus active use, so we asked: **is more daily social media use associated with a lower self-rated mental health score, and how much of that link survives other explanations?**

Data and approach

We used the *Students' Social Media Addiction* dataset (Kaggle): 705 self-reported survey responses, 13 variables. After removing one duplicated row, **n = 704** students from **110** countries remained. We fitted a simple linear regression with a two-sided t-test and 95% confidence interval, checked predictions on held-out students and held-out countries, and then tested competing explanations: country clustering, platform type, sleep as a pathway, double machine learning (DML) to remove measured background differences, and a sensitivity analysis for unmeasured confounders.

Key findings

Finding	Evidence
Heavier use goes with lower mental health	−0.70 points per extra hour (95% CI −0.74 to −0.66; t = −35.5, p < 0.001); R^2 = 0.64. A rank correlation gives the same answer (Spearman ρ = −0.80).
Predictions hold for held-out students and countries	Test R^2 0.64 across 200 random splits (range 0.48–0.73) and 0.62 when whole countries are held out; error 0.66 vs 1.10 for guessing the mean. Checked within this dataset only. For one student the 95% prediction interval is about ±1.3 points.
Country matters about as much as hours	Country alone explains 63% of score variation after adjustment (69% raw), similar to hours (64%); together 82%. Within countries the slope is −0.58 (country-clustered 95% CI −0.73 to −0.42).
Platform type changes the slope	Slopes range from −0.41 (WhatsApp) to −1.05 (Twitter); one common slope is rejected, F(8, 650) = 3.59, p < 0.001, and all 9 platform slopes survive a Bonferroni correction. Platform is only a proxy for purpose of use.
Sleep carries part of the link	About 20% runs through sleep (indirect effect −0.14, 95% CI −0.18 to −0.10).
The link survives measured background	After DML removes age, gender, academic level, relationship status and country: −0.63 per hour (95% CI −0.68 to −0.59; this interval ignores country clustering). A hidden confounder would need to explain at least 71% of the remaining variation in both hours and scores to erase the simple-regression slope.

What the data actually shows

The conventional wisdom points in the right direction but tells the wrong story. Hours matter, roughly 0.6 points per hour even after removing measured background, but they are tangled up with where students live, how much they sleep and which platform they use. Because the data is cross-sectional and self-reported, it cannot tell us whether heavy use lowers mood or low mood drives heavy use, and unmeasured stress could drive both.

Limitations

The sampling frame is undocumented and 79 of 110 countries (72%) contribute a single student, so results describe this dataset rather than all students. The data looks unusually clean: the addiction score correlates −0.95 with mental health, within each of three platforms every user gives the same score, and "social media hurts my studies" answers jump from 0% to 100% across usage bands. Stress and purpose of use were never measured. The data is public and de-identified, but its consent process is undocumented. We treat the results as a teaching example, not as evidence for policy.

Recommendations

- **Counsellors:** treat heavy use as a reason to start a conversation, not as a diagnosis, and ask about sleep early.
- **Students:** for one week, note daily hours and sleep; if sleep drops on heavy-use days, protecting sleep is a concrete first step.
- **Researchers:** add purpose of use, a validated stress scale and phone-recorded screen time; sample at least 30 students per country and follow them for a semester.